Location Prediction Using Bayesian Optimization LSTM for RIS-Assisted Wireless Communications

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***Abstract*—Reconfigurable intelligent surface (RIS) represent a novel form of electromagnetic metamaterial that have been extensively studied for user equipment (UE) positioning by exploiting the multipath propagation of signals. A novel RIS-assisted localization prediction (RLP) method based on Bayesian optimization and long short-term memory (BO-LSTM) has been proposed in this paper. This method capitalizes on the predictive advantages of LSTM for data sequence and RIS's flexible and controllable multidimensional feature parameters, establishing a mobile UE localization model in an RIS-assisted wireless communications system based on the interplay between time slot transmission power and user location information. In order to provide a more stable communication environment for data collection during the localization process, a power allocation optimization (PAO) method is proposed for maximizing time slot channel capacity in the RLP system based on the number of RIS reflection elements. The study conducts a thorough comparison of simulation results of BO-LSTM, convolutional neural networks (CNN)-LSTM and improved bidirectional LSTM (BiLSTM) combined with Adaptive boost, employing adaptive moment estimation (Adam) and stochastic gradient descent with momentum (SGDM) optimizers. Experimental results demonstrate that the BO-LSTM-based RLP method exhibits improved prediction accuracy. These findings suggest the effectiveness of the proposed method and highlight its potential for further enhancement.**

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***Index Terms*—Reconfigurable intelligent surface, Bayesian optimization LSTM, location prediction, wireless communications.**

I. Introduction

RECONFIGURABLE intelligent surface (RIS)-assisted communication system is an emerging communication technology with revolutionary potential [1], [2]. It has garnered widespread attention for its outstanding information exchange capabilities and applications in the field of simultaneous transmission and reception. Location-based information exchange plays a crucial role in the research and application of RIS-assisted wireless communication systems, as it determines how RIS collaborated with other communication elements to provide an optimal communication experience [3]. With the continuous development of RIS technology, location-based information exchange capabilities have become a key factor in achieving more intelligent, sustainable, and efficient communication systems.

Owing to the involvement of numerous characteristic parameters in the process of RIS-assisted communication transmission, such as phase modulation, frequency selection, and the dynamic adjustment of these parameters based on the environment and communication requirements, leveraging the features of RIS can lead to enhanced performance and precision in communication localization issues [4]. This is particularly beneficial in a dynamically fast-changing communication and positioning environments, and the flexibility and controllability of RIS make it a highly regarded technology in the field of wireless communication localization. RIS-assisted positioning offers another advantage by facilitating localization functionality in asynchronous wireless communication systems [5]. Addressing nonlinear optimization challenges inherent in asynchronous positioning, the utilization of suitable deep learning (DL) optimization algorithms can effectively mitigate errors and yield more precise positioning outcomes.

6th generation mobile communication system (6G) aims to further propel the development of smart cities, with centimeter-level precision in communication sensing and positioning being a crucial focus in 6G technology. DL has emerged as a vital tool for aiding in predicting the locations and movements of user equipment (UE) within communication systems. In particular, the long short-term memory (LSTM) [6] algorithm within DL [7], due to its structure allowing parallel computation during training, exhibits faster convergence speed, making it more suitable for applications in communication localization scenarios.

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Additionally, its adaptability to various data types and tasks makes it advantageous in predicting a user's trajectories and interactive modeling of information, showing superiority over traditional machine learning (ML) algorithms [8].

Combining DL with RIS-assisted communication localization scenarios is a promising positioning method that can effectively enhance the accuracy and reliability of localization. Utilizing DL technology enables the learning of characteristic parameters in the RIS-assisted communication localization system and the establishment of feature models for communication signals. These models serve to extract target positions and environmental data, optimizing signal transmission and discerning critical location details [9]. In the RIS-assisted communication localization process, the ability to jointly estimate the time of arrival (TOA), angle of arrival (AOA), and time difference of arrival (TDOA) of different signals enhances accuracy. Consequently, this facilitates dynamic adjustments to training data in DL, thereby improving the accuracy of localization information.

The application value of the Bayesian optimization (BO)-LSTM assisted RIS communication localization algorithm proposed on realizing high-precision prediction of mobile user's position. In this approach, LSTM leverages the parameters of the RIS as input characteristic data, enhancing the flexibility of the localization algorithm, thereby enabling its applicability in diverse and complex communication scenarios. Given the dynamic nature of user communication states in a communication environment, one of the advantages of LSTM is its adaptability to real-time and dynamic data. This capability allows for the real-time adjustment of localization strategies, ensuring the maintenance of efficient communication performance.

A. Motivation and Contributions

Due to the existing research on RIS-assisted communication localization methods not fully harnessing the advantages of DL, there is an opportunity to enhance localization accuracy by utilizing various performance parameters of RIS as multi-dimensional feature data in DL training data. As a controllable intelligent device, RIS can provide substantial and effective information data containing multi-dimensional features as feedback. This innovation in space offers a novel perspective for DL-based localization problems. In the process of RIS-assisted localization, where signal propagation involves long-distance dependencies, LSTM proves effective in handling dependencies among long sequences. Additionally, LSTM adeptly captures and utilizes temporal information in the data, adapting to dynamically changing channel conditions and RIS configurations. Consequently, it addresses key challenges in the RIS localization process.

This paper presents a solution to the problem of predicting the location of a UE in a RIS-aided communication scenario by training on prior UE power data. Our approach combines the LSTM algorithm in machine learning with BO. The main contributions include:

- Leverages the advantages of the LSTM algorithm in data sequence prediction to establish a mobile user localization method based on the relationship between time-slot transmission power and its location in an RIS-assisted wireless communication system.
- Proposes an algorithm for optimizing the power allocation of time slot channel capacity based on the number of reflecting elements on the RIS in the RIS-assisted localization system strategy. The optimized power allocation is employed in LSTM assisted localization.
- Utilizes RIS characteristic parameters to provide multi-dimensional characteristic data for the LSTM localization prediction model. The flexible controllability of RIS characteristic parameters makes the proposed localization algorithm applicable to diverse scenarios.
- Employs BO as a data preprocessing module for LSTM assisted RIS localization system. The reasons for the improvement in prediction accuracy for LSTM-assisted RIS communication localization by analyzing the relationship between posterior and prior probabilities in BO is deduced. The accuracy of the proposed method is verified in Section V, and it is compared with CNN as a data preprocessing method, showing that BO has higher accuracy.
- An improved BiLSTM-Adaptive boost (Adaboost) algorithm is designed on the basis of the advantages of LSTM algorithm for time series prediction. Adaboost obtains different weak predictors by updating the weight of each round of data training, and integrates all the predictors to obtain the final combined predictor. The prediction results obtained can prove the effectiveness of the improved method.

B. Organizations and Notations

The remainder of the paper is organised as follows. Section II introduces the related works, and Section III shows the system model, including the scenario model, the channel model, and the problem formulation analysis. The algorithm procedure of RIS-assisted BO-LSTM location prediction method and network architecture is described in Section IV. Section V shows the simulation results. Section VI concludes with subsequent thoughts and future directions on the scheme and summarizes the framework of the proposed approach.

Notations: $\mathbb{C}^{m\times n}$ denotes the space of $m \times n$ real (complex) matrices. $\mathbb{R}^d$ expresses the set of real numbers of length d. $\mathbf{H}$ and H are general matrix. $\mathrm{diag}(\mathbf{x})$ represents the diagonal matrix with vector $\mathbf{x}$ for its diagonal elements. $\|\cdot\|$ stands for the second order Euclidean norm, and $\arg\max \alpha_m(\tilde{x})$ presents the parameter $\tilde{x}$ in function $\alpha_m(\tilde{x})$ that produces the maximum output. The distribution of the circularly symmetric complex Gaussian (CSCG) with zero mean and variance γ^2 is denoted by $\mathcal{CN}(0,\gamma^2)$. ∇ denotes the gradient operation. Finally, $\mathcal{N}(,)$ denotes the normal distribution.

II. Related Work

RIS optimization is a promising approach to improve the effectiveness of positioning systems. RIS assisted the TDOA algorithm for locating radio-frequency (RF) transmitter proposed by [11], which improved the effectiveness of traditional

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positioning methods. Joint optimization of the RIS phase distribution and BS active precoding localization and synchronization method [12] can reduce the computational complexity, and improve the robustness of the scheme.

Several recent studies, such as references [13], [14], [15], have underscored the effectiveness of LSTM neural networks for location estimation, consistently achieving high accuracy results. In [16], a hybrid Markov model was proposed that integrates with an LSTM model for location prediction, outperforming five existing baseline methods in terms of predictive performance. Another approach, presented in [17], used a deep convolutional neural network (CNN)-LSTM architecture for passive infrared (PIR)-based indoor location estimation, while [18] proposed a spatio-temporal position prediction model (SPPM) based on LSTM in mobile edge computing (MEC). [19] used a trained LSTM network to predict the source location of sound, and numerical simulations demonstrate the effectiveness of LSTM for underwater source localization. [20] proposed a hybrid precoding architecture based on geometric mean decomposition (GMD) that jointly considered the design of the LSTM autoencoder scheme, while [21] introduced a novel LSTM-based scheme that uses dynamic information about the wireless network and energy harvesting of RIS to make decisions.

As a variant of recurrent neural network (RNN), LSTM has been commonly employed for data sequence modeling and research in localization techniques. Based on historical trajectory data, it can also be used to predict the future positions of users. Due to the relatively complex environment of indoor wireless location estimation, [13] proposed a low-power indoor positioning method for Bluetooth devices based on an end-to-end LSTM neural network. [22] introduced a location prediction model based on LSTM by considering wireless measurement reports from serving and neighboring base stations (BS). [23] combined a CNN based on visual feature extraction with a bidirectional LSTM network incorporating a small inertial measurement unit. In [24], a variant of LSTM-based recurrent units was designed and experimentally validated on two real semantic trajectory datasets. [25] optimized sensor layout using an LSTM model, considering various indoor fire scenarios to create a simulation database for real-time fire localization. [26] demonstrated that LSTM networks can automatically construct measurement-based scalable information vectors by leveraging potential temporal correlations between measurements. Furthermore, DL can optimize RIS orientation and phase shifts, as evidenced in [27]. Through simulation and analysis, this study identified optimal RIS deployment locations and proposed a scheme to minimize computational complexity. However, these method relying on signal strength differences for positioning might be sensitive to environmental factors such as interference, obstacles, or changes in the wireless communication environment, affecting the accuracy of location estimation.

BO has found extensive application in DL research. Employing data preprocessing techniques alongside BO can enhance the accuracy of LSTM models while minimizing the number of layers required to achieve optimal performance. For example, [28] proposed an approach to predict changes in the ON-state resistance of MOSFETs using a BO-LSTM RNN, which demonstrated improved accuracy compared to traditional LSTM models. Similarly, BO-LSTM has also been applied to predict wind speed [29] and power load of large public buildings [30], leading to significant improvements in accuracy.

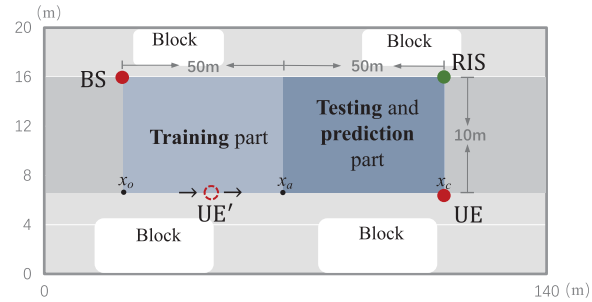

Fig. 1. RIS-aided location prediction model.

The innovative applications combining LSTM with RIS have brought smarter and more adaptive efficient solutions to the wireless communication domain. In order to meet the data rate and consistency requirements of 6G, [31] proposed a channel estimation scheme utilizing CNN-LSTM to assist the RIS-NOMA system. The accuracy of this channel estimation method was evaluated through various performance metrics. To improve energy efficiency, [32] proposed a scheme utilizing LSTM to leverage dynamic information from wireless networks and make decisions based on channel complexity and RIS energy harvesting. Due to the difficulty of RIS addressing the mixed precoding problem, [33] introduced a design using a combination of GMD and LSTM in an autoencoder scheme. In RIS-assisted short-packet communication (SPC) systems, [34] proposed a DL method to optimize the number of reflective elements in each RIS or the number of RISs in a set of distributed RISs. The research findings indicate that deep neural network (DNN) can save costs while meeting communication requirements. An application of LSTM in RIS-assisted intelligent transportation systems is presented in [35]. In these researches, issues related to the computational complexity and resource requirements for scaling up the proposed solutions could be potential limitations.

The optimization study of RIS-assisted wireless communication channel capacity involves more efficient allocation of spectrum resources [36], contributing to improved signal distinguishability and localization accuracy, as well as enhanced management of mobile device location updates. Utilizing DL approach, [37] jointly optimized RIS phase shifts and drone trajectories, leading to increased channel capacity in non-orthogonal multiple access (NOMA) networks. Addressing edge caching constraints and power limitations, [38] proposed a joint optimization problem for power allocation based on collaborative caching, demonstrating its effectiveness in reducing power consumption.

III. System Model

A. System Model of Location Prediction

The system model of the proposed RIS-aided location-prediction (RLP) method is shown in Fig. 1. The system

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TABLE I
NOTATIONS THROUGHOUT THIS PAPER

Notation	Definition	Notation	Definition
k	Number of reflection elements in RIS.	$\bar{N}$	Number of transmission antennas of BS.
H_{sd}	Channel matrices for BS-UE.	$\widetilde{\mathbf{H}}_{sr,k}$	Channel matrices for BS-RIS.
$\widetilde{\mathbf{H}}_{rd,k}$	Channel matrices for RIS-UE.	s	Transmit signal.
$\hat{\widetilde{\mathbf{H}}}^{l}_{sr,k}$	The estimated temporal channel matrices for BS-RIS.	$\hat{\widetilde{\mathbf{H}}}^{l}_{rd,k}$	The estimated temporal channel matrices for RIS-BS.
$\hat{H}^{l}_{sd}$	The estimated temporal channel matrices for BS-UE.	$\widetilde{\mathbf{H}}^{l}_{sr,k}$	The channel matrices for BS-RIS at the l-th position.
$\widetilde{\mathbf{H}}^{l}_{rd,k}$	The channel matrices for RIS-UE at the l-th position.	H^{l}_{sd}	The channel matrices for BS-UE at the l-th position.
n_o	The noise of receiver.	γ^2	The power of noise.
θ_k	The angle of incidence wave between BS and UE.	ϑ_k	The angle of departure wave between RIS and UE.
ϱ_k	The amplitude reflection coefficient.	φ_k	The phase-shift variable.
$p^l[i]$	Transmission power in i-th time slot.	$x[i]$	Distance from initial position to current position.
α_k	Power distribution coefficient.	B	Bandwidth.
$\widetilde{x}[m]$	Hyperparameter for BO model.	μ_k	Lagrange multiplier.
μ_1	Hyperparameter which controls first-order momentum.	μ_2	Hyperparameter which controls second-order momentum.
λ_k	Lagrange multiplier.	r_{opt}	The decision rule function of GPR.
r	Evaluated true value vector of GPR.	r_*	Global maximum value of GPR.
q	True distribution of BO posterior distribution.	$\widetilde{r}_*$	Local maximum value of $f(\widetilde{x}[m])$.
β	Shadow factor.	β_i	Initial learning rate.
ε_p	The error of prediction data.	σ_p	The variance of error ε_p.
J	The length of estimated time series data.	τ	The total length of past time series data.
$\tilde{X}$	The collection of initialized samples.	ζ	Achievable rate.
χ	The hyperparameter exploration space.		

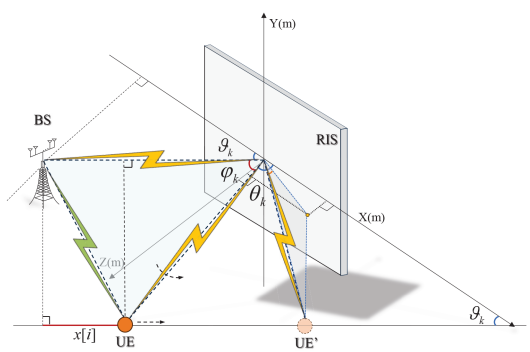

Fig. 2. Schematic representation of characteristic data based on RIS.

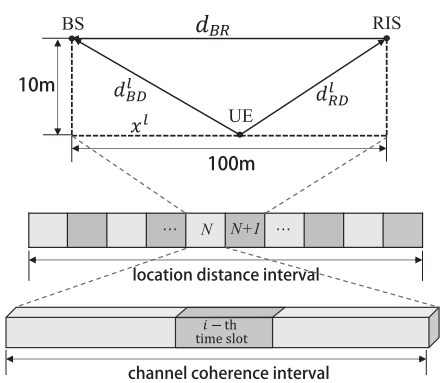

Fig. 3. TDD system channel model for proposed RLP system.

comprises a BS and a RIS that are fixed, while a UE moves along a straight line. Considering a multiple input multiple output (MIMO) scenario [39], where the BS has $\bar{N}$ antennas, the RIS consists of k elements that support an uplink communication system. Let $\mathbf{H}_{sr,k}$ and $\mathbf{H}_{rd,k}$ denote the BS-RIS and RIS-UE channel coefficient matrices, respectively. The transmission path includes a direct link between the UE and BS, as well as a cascade link of UE-RIS-BS. The received signal vector at the UE is denoted as [40]:

$$y = \sum_{k=1}^{K} \sqrt{p}\left(H_{sd} + \widetilde{\mathbf{H}}_{sr,k}\Theta\widetilde{\mathbf{H}}_{rd,k}\right)s + n_o, \tag{1}$$

where $\widetilde{\mathbf{H}}_{sr,k} = \mathbf{H}_{sr,k}e^{-j\theta_k} \in \mathbb{C}^{k\times\bar{N}}$, $\widetilde{\mathbf{H}}_{rd,k} = \mathbf{H}_{rd,k}e^{-j\vartheta_k} \in \mathbb{C}^{k}$ and $H_{sd} \in \mathbb{C}^{\bar{N}}$ represent the channel matrices of BS-RIS, RIS-UE, and BS-UE, respectively, following the Rayleigh distribution. θ_k and ϑ_k, shown in Fig. 2, signify the angle of incidence wave between BS and UE and departure wave between RIS and UE. The diagonal matrix $\Theta = \mathrm{diag}(\varrho_1 e^{j\varphi_1}, \varrho_2 e^{j\varphi_2}, \ldots, \varrho_k e^{j\varphi_k})$, denotes the phase shift for every element of RIS, in which ϱ_k and $\varphi_k \in [0, 2\pi]$ are the amplitude reflection coefficient and the phase-shift variable, respectively. The transmit power and signal from BS are depicted as $\sqrt{p}$ and s. The noise of the receiver is expressed as $n_o \sim \mathcal{CN}(0, \gamma^2)$ with zero mean and variance γ^2.

B. TDD System for RIS-Aided Uplink Channel

In the RIS-assisted localization prediction (RLP) system model, it is assumed that the system adopts a time-division duplex (TDD) communication mode, and pilots are allocated to different positions of UE in the uplink. As Fig. 3 shows, the BS uses the reciprocity of the channel to obtain the CSI of the uplink channel estimation from UE to BS for the precoding matrices of the BS downlink channel and uses the obtained information for data subsequent prediction. This scenario is selected from the Nth section and $(N+1)^{\text{th}}$ section in a long location distance interval. The former part is used for statistic model training, while the latter part is used for prediction and accuracy calculation. Based on the time correlation demonstrated in the channel modeling by the channel estimation in [35] and [41], the specific

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formula of channel time correlation is:

$$\hat{H}^l_{sd}[i+1],\ldots,\hat{H}^l_{sd}[i+J]=f_e\left(H^l_{sd}[i-\tau+1],\ldots,H^l_{sd}[i]\right), \tag{2}$$

$$\hat{\tilde{\mathbf{H}}}^l_{sr,k}[i+1],\ldots,\hat{\tilde{\mathbf{H}}}^l_{sr,k}[i+J]=f_e\left(\tilde{\mathbf{H}}^l_{sr,k}[i-\tau+1],\ldots,\tilde{\mathbf{H}}^l_{sr,k}[i]\right), \tag{3}$$

$$\hat{\tilde{\mathbf{H}}}^l_{rd,k}[i+1],\ldots,\hat{\tilde{\mathbf{H}}}^l_{rd,k}[i+J]=f_e\left(\tilde{\mathbf{H}}^l_{rd,k}[i-\tau+1],\ldots,\tilde{\mathbf{H}}^l_{rd,k}[i]\right), \tag{4}$$

where $\tilde{\mathbf{H}}^l_{sr,k}[i]$, $\tilde{\mathbf{H}}^l_{rd,k}[i]$ and $H^l_{sd}[i]$ denote the channel matrices from BS to RIS, from RIS to UE, and the direct link from BS to UE in the i-th time slot at the l-th position. $\hat{H}^l_{sd}[i]$, $\hat{\tilde{\mathbf{H}}}^l_{sr,k}[i]$ and $\hat{\tilde{\mathbf{H}}}^l_{rd,k}[i]$ denote the estimated temporal channel matrices for BS-UE, BS-RIS and RIS-UE in the i-th time slot at the l-th position. J and τ are the length of estimated time series data and the total length of past time series data, respectively. $f_e(\cdot)$ represents an arbitrary mapping function that needs to be predicted. The received signal vector by the UE in the i-th time slot at the l-th position within a channel coherence interval is expressed as:

$$y^l[i]=\sum_{k=1}^{K}\sqrt{p^l[i]}\left(H^l_{sd}[i]+\tilde{\mathbf{H}}^l_{sr,k}[i]\Theta^l[i]\tilde{\mathbf{H}}^l_{rd,k}[i]\right)s^l[i] + n^l_o[i], \qquad i=0,1,2,\ldots,I-1, \tag{5}$$

where $p^l[i]$, $s^l[i]$, $\Theta^l[i]$ and $n^l_o[i]$ are the transmit power, the signal from BS, the phase shift matrix for every element of RIS and noise in the i-th time slot at the l-th position.

In the i-th time slot of the user at l-th position, the pilot signal received by the BS is as follows:

$$\begin{aligned}F^l_{BS}[i]&=\beta\sum_{k=1}^{K}y^l[i]\sqrt{P^l_{UE}[i]}+n^l[i]\\&=\beta\sum_{k=1}^{K}s^l[i]\left(H^l_{sd}[i]+\widetilde{\mathbf{H}}^l_{sr,k}[i]\Theta^l[i]\widetilde{\mathbf{H}}^l_{rd,k}[i]\right)\sqrt{p^l[i]P^l_{UE}[i]}\\&\quad+n^l_o[i], \qquad i=0,1,2,\ldots,I-1,\end{aligned} \tag{6}$$

where $\beta=(\frac{\lambda}{d^l_{BD}[i]})^v$ represents the fading factor [42], $d^l_{BD}[i]$ represents the distance of BS to UE, v and λ is the path loss exponent and wavelength. $P^l_{UE}[i]$ denotes transmit power of UE. In the uplink transmission system, the pilot signal from the UE needs a certain level of transmission power so that the BS can receive the pilot signal depite the effects of path loss, and the transmission power is denoted by:

$$\begin{aligned}P^l_{BS}[i]&=P^l_{UE}[i]A_{UE}A_{BS}\left[\frac{\lambda}{4\pi d^l_{BD}[i]}\right]^2\\&\quad+P^l_{UE}[i]A_{UE}A_{BS}\left[\frac{\lambda}{4\pi\left(d_{BR}+d^l_{RD}[i]\right)}\right]^2,\end{aligned} \tag{7}$$

where A_{UE} and A_{BS} denote the antenna gain of UE and BS, the distance from RIS to UE and UE to BS are presented as $\|d^l_{RD}[i]\|=\sqrt{d^2_{BD,\min}[i]+(d_{BR}-x[i])^2}$ and $\|d^l_{BD}[i]\|=\sqrt{(x[i])^2+d^2_{BR}}$, where $x[i]$ is the distance from the starting point to l-th position and d_{BR} express the distance between BS and RIS which is a constant value, $d_{BD,\min}$ indicates the shortest distance between the BS and the UE. The channel capacity of RLP system can be expressed as:

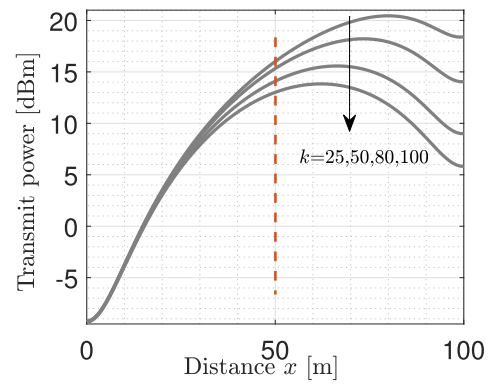

Fig. 4. Visualization of transmit power curves for different numbers of elements in RIS ($k=25$, 50, 80, 100).

$$C_{RIS,k}[i]=\max_{\varphi_1,\ldots,\varphi_k}\log_2\left(1+SNR_{RIS}[i]\right), \tag{8}$$

where

$$SNR_{RIS}[i]=\frac{p^l[i]\left|H^l_{sd}[i]+\tilde{\mathbf{H}}^l_{sr,k}[i]\Theta[i]\tilde{\mathbf{H}}^l_{rd,k}[i]\right|^2}{\gamma^2}, \tag{9}$$

denotes the signal-to-noise ratio (SNR) of UE in the i-th time slot at the l-th position in a channel coherence interval. The transmit power needed for the RIS-aided location system for the UE will be:

$$p^l[i]=\left(2^\zeta-1\right)\frac{\gamma^2}{\left|H^l_{sd}[i]+\widetilde{\mathbf{H}}^l_{sr,k}[i]\Theta^l[i]\widetilde{\mathbf{H}}^l_{rd,k}[i]\right|^2}, \tag{10}$$

where ζ expresses the achievable data rate required for the system. The transmit power function is visualized in Fig. 4 as numerically obtained by (7). The data sequence obtained through the model in Fig. 1 can be further stacked into:

$$\mathrm{P}^l=\left\{p^l[0],p^l[1],\ldots,p^l[i]\right\} \quad i=0,1,\ldots,I-1, \tag{11}$$

where $p^l[i]$ represents the transmission power associated with the i-th time slot, and the data of transmission power can be represented in sequence as P^l. In different time slots, there will be different corresponding transmission power values.

C. Characteristic Data of RIS-Aided Location System

RIS-assisted wireless communication scenarios can introduce more location-related characteristic parameters in the prediction of positioning data, i.e., incident signal angle θ_k, phase shift angle φ_k and departure signal angle ϑ_k, which can greatly improve the accuracy and robustness of the localization model. Therefore, in the proposed LSTM-assisted wireless communication localization scenario, RIS is used to provide these multi-dimensional

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features for the localization scene to compensate against noise in the data. Let $\{\theta_k\}_i^{I-1}$, $\{\varphi_k\}_i^{I-1}$ and $\{\vartheta_k\}_i^{I-1}$ be the set which covers the potential space of incident signal angle, phase shift angle and departure signal angle, as shown in Fig. 2. The discrete signal direction set can be expressed as:

$$\begin{aligned}\mathbf{\Phi}_{RIS} = [&\mathbf{Q}_1(\{\theta_k\}_1,\{\varphi_k\}_1,\{\vartheta_k\}_1),\ldots,\\ &\mathbf{Q}_i(\{\theta_k\}_i,\{\varphi_k\}_i,\{\vartheta_k\}_i),\\ &\ldots,\mathbf{Q}_{I-1}(\{\theta_k\}_{I-1},\{\varphi_k\}_{I-1},\{\vartheta_k\}_{I-1})].\end{aligned} \tag{12}$$

The correlation between these angle-related feature parameters and each quantity in Fig. 3 are as follows according to the law of cosines:

$$\begin{cases}2d_{BR}d_{RD}^l[i]\cos\{\varphi_k\}_i^{I-1} = d_{BR}^2 + (d_{RD}^l[i])^2 - (d_{BD}^l[i])^2,\\ d_{RD}^l[i]\cos\{\varphi_k\}_i^{I-1} = d_{BR} - x[i],\end{cases} \tag{13}$$

$$\begin{cases}2d_{RD}^l[i]d_{BD,\min}\sin(\{\vartheta_k\}_i^{I-1} + \{\theta_k\}_i^{I-1})\\ = (d_{RD}^l[i])^2 + d_{BD,\min}^2 - (d_{BR} - x[i])^2,\\ d_{BD,\min}\sin(\{\vartheta_k\}_i^{I-1} + \{\theta_k\}_i^{I-1}) = d_{RD}^l[i].\end{cases} \tag{14}$$

According to the existing feature quantities $\{\theta_k\}_i^{I-1}$, $\{\varphi_k\}_i^{I-1}$ and $\{\vartheta_k\}_i^{I-1}$, $x[i]$ containing multiple feature quantities can be expressed as:

$$x[i] = [x_{\theta_{ki},i}, x_{\varphi_{ki},i}, x_{\vartheta_{ki},i}], \tag{15}$$

where

$$x_{\theta_{ki},i} = x(\{\theta_k\}_i), x_{\varphi_{ki},i} = x(\{\varphi_k\}_i), x_{\vartheta_{ki},i} = x(\{\vartheta_k\}_i), \tag{16}$$

express the interaction between location information and angle information of RIS.

Therefore, the location information sequence $x[i]$, containing all angle feature parameters, can be represented as a multidimensional matrix:

$$X = \begin{bmatrix} x_{\theta_{k1},1} & x_{\theta_{k2},2} & \ldots & x_{\theta_{ki},i} & \ldots & x_{\theta_{k(I-1)},I-1}\\ x_{\varphi_{k1},1} & x_{\varphi_{k2},2} & \ldots & x_{\varphi_{ki},i} & \ldots & x_{\varphi_{k(I-1)},I-1}\\ x_{\vartheta_{k1},1} & x_{\vartheta_{k2},2} & \ldots & x_{\vartheta_{ki},i} & \ldots & x_{\vartheta_{k(I-1)},I-1}\end{bmatrix}_{3\times(I-1)}. \tag{17}$$

Finally, the relation of the transmission power sequence of P^l corresponding to location information X is denoted by:

$$\mathrm{P}^l(X) \sim \left\{p^l(x[0]),\ldots,p^l(x[i]),\ldots,p^l(x[I-1])\right\}. \tag{18}$$

Each location information $x[i]$ corresponds to a transmission power $p^l(x[i])$. Thus, $\mathrm{P}^l(X)$ represents a sequence of transmission power $p^l(x[i])$, which reflects the relationship between location information X and transmission power, and also reflects the effect of transmission power on location information.

D. The Power Allocation Optimization Problem in RLP System

In the proposed RIS-assisted localization problem, power allocation and channel capacity optimization are interrelated. Considering the reflection characteristics of RIS and the total transmission power at each time slot, different transmission power is allocated to each reflection element on RIS to maximize the channel capacity. Through this optimization, the transmission and reception of signals are improved to maximize the reliability and accuracy of positioning information. As a result, the optimization of RIS-assisted localization is formulated as the maximization of the channel capacity in the i-th time slot subject to constraints on the total power and the power allocation factor of each reflecting element of the RIS as follows:

$$\max\quad C_{\text{sum},RIS}[i] = \sum_{k=1}^{K} B\log_2\left[\det\left(1 + \frac{P_{\text{sum}}^l(X)}{\gamma^2}\right.\right.$$
$$\left.\left.\times\left\|H_{sd,k}^l[i] + \tilde{\mathbf{H}}_{sr,k}^l[i]\Theta^l[i]\tilde{\mathbf{H}}_{rd,k}^l[i]\right\|\right)\right], \tag{19a}$$

$$\text{s.t.}\quad \text{C1}: P_{\text{sum}}^l(X) = \sum_{k=1}^{K}\alpha_k p^l(x[i]), \tag{19b}$$

$$\text{C2}: \sum_{k=1}^{K}\alpha_k = 1,\quad \alpha_k \in [0,1], \tag{19c}$$

$$\text{C3}: p^l(x[i]) > p_{\min}^l(x[i]),\quad l = 1,2,\ldots,L, \tag{19d}$$

$$\text{C4}: x[i] \in [x_a, x_c],\quad i = 0,1,\ldots,I-1, \tag{19e}$$

$$\text{C5}: \forall x_a \in \mathcal{A}, \forall x_c \in \mathcal{C}, \tag{19f}$$

where α_k in C1 represents the power distribution coefficient for each reflection element on RIS, $p_{\min}^l(x[i])$ denotes the minimum transmit power requirement of UE, $\mathcal{A}$ and $\mathcal{C}$ represent the range of values of x_a and x_c in meters ($x_c > x_a$), B stands for bandwidth. The minimum transmission power point within the constraint range of x_a to x_c is obtained by the sequence of performance parameters from the initial location value x_o to x_a of the model. The performance of the RLP system, i.e., the transmission power required by the terminal, reflects the corresponding relationship between the terminal position.

IV. Network Architecture of BO-LSTM Localization Prediction Algorithm

The RIS-assisted BO-LSTM location prediction method combines the use of BO-LSTM with RIS-assisted communication localization prediction algorithm (BL-RLP). The algorithm procedure involves initializing the dataset and training the LSTM model using known *a-priori* location data. Then, BO is used to optimize the hyperparameters of the LSTM model and obtain the relationship of transmit power and position of UE.

The primary objective of the proposed RIS-assisted localization prediction method based on BO-LSTM as shown in Fig. 5 is to determine the optimal parameters of the proposed RLP model through training sample data, specifically, the parameter estimation of the model. The symbols and meanings in Fig. 5 are shown in Table II. In the RIS-aided communication model in this paper, the transmission power of the UE is predicted in the next segment location by training the model obtained from the transmission power data required by UE in a segment of a known location. The prediction model can be used to determine the relationship between the transmission power required by

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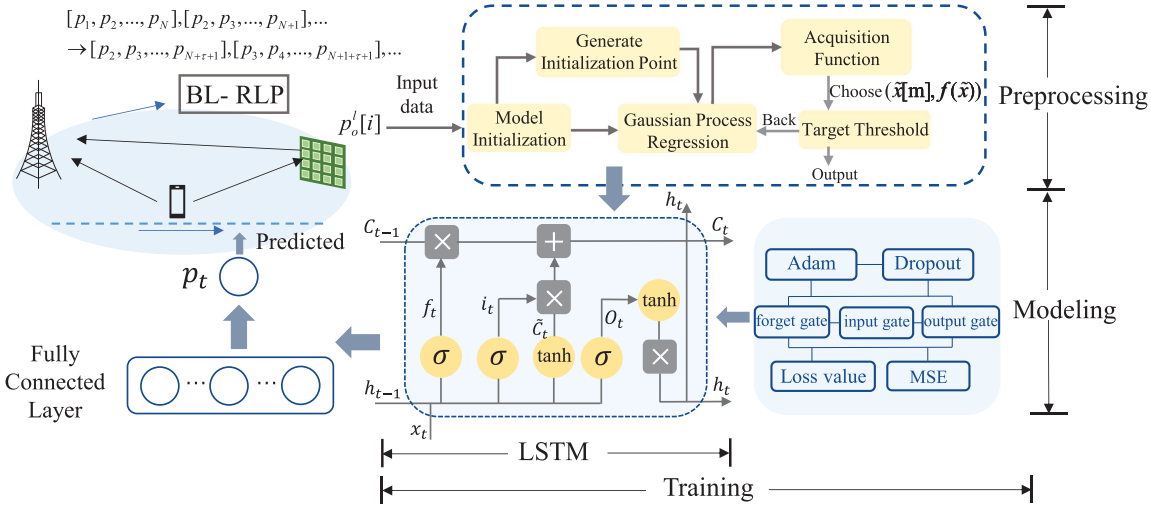

Fig. 5. Network architecture of BO-LSTM algorithm for proposed localization model.

TABLE II
SYMBOL AND MEANING OF BO-LSTM NETWORK ARCHITECTURE

Symbol	Meaning	Symbol	Meaning
i_t	The output signal of the input gate at time t	σ	Sigmoidal activation function
$\tilde{C}_t$	Candidate value vector at time t	tanh	Hyperbolic tangent function
O_t	Output signal in output gate at time t	f_t	Output signal of forget gate at time t
C_t	Memory unit at time t	x_t	Input sequence data at time t
C_{t-1}	Memory unit at time $t-1$	h_t	Implicit state at time t
b	Hidden and output layer neuron thresholds	h_{t-1}	Implicit state at time $t-1$
W	Weight of convolution operation		

UE and its location and predict the location of UE when the transmission power required is the minimum as shown in Fig. 4.

A. The Power Allocation Optimization Method in RLP System

In this section, the optimization method for power allocation (PAO) in i-th time slot is presented. Transmission power levels are allocated to individual reflecting elements based on the varying number of RIS elements, aiming to maximize channel capacity. Maximizing capacity is essential for enhancing the robustness and accuracy of the localization scenario. Specifically, a larger channel capacity contributes to higher channel stability, increased stability in receiving and feedback of signals from users, and is advantageous for the collection and computation of location-related data. In the context of the problem formulation in (19), the objective function $C_{\text{sum},RIS}[i]$ can be reformulated as:

$$\begin{aligned} C_{\text{sum},RIS}[i] &= \sum_{k=1}^{K} B\log_2\left[1 + p^l(x[i]) \cdot \frac{\alpha_k \left\|\tilde{\mathbf{H}}^l_{RIS,k}[i]\right\|}{\gamma^2}\right] \\ &= \sum_{k=1}^{K} B\log_2\left[\frac{\alpha_k \left\|\tilde{\mathbf{H}}^l_{RIS,k}[i]\right\|}{\gamma^2} + p^l(x[i])\right] \\ &\quad + \sum_{k=1}^{K} B\log_2\left[\frac{\gamma^2}{\alpha_k \left\|\tilde{\mathbf{H}}^l_{RIS,k}[i]\right\|}\right]. \end{aligned} \tag{20}$$

To simplify notation for computational convenience, define $\tilde{\mathbf{H}}^l_{RIS,k}[i] = H^l_{sd,k}[i] + \tilde{\mathbf{H}}^l_{sr,k}[i]\Theta^l[i]\tilde{\mathbf{H}}^l_{rd,k}[i]$. From the results obtained in (20), we can summarize a quadratic programming problem related to the independent variable $p^l(x[i])$, and express it as:

$$\begin{aligned} \max \quad & \sum_{k=1}^{K} B\log_2\left[\frac{\alpha_k \left\|\tilde{\mathbf{H}}^l_{RIS,k}[i]\right\|}{\gamma^2} + p^l(x[i])\right], \\ \text{s.t.} \quad & P^l_{\text{sum}}(X) = \sum_{k=1}^{K} \alpha_k p^l(x[i]). \end{aligned} \tag{21}$$

The problem is evidently a convex optimization problem, and the slater condition is satisfied, i.e., there exists $p^l(x[i])$ such that $\sum_{k=1}^{K} \alpha_k p^l(x[i]) - P^l_{\text{sum}}(X) \leq 0$ holds. In this case, the Karush-Kuhn-Tucker (KKT) [43] conditions are necessary and sufficient for optimality. Introducing Lagrange multipliers λ_k and μ_k, the Lagrange function is expressed as:

$$\begin{aligned} \mathcal{L}(p^l(x[i])) &= -\sum_{k=1}^{K} B\log_2\left[\frac{\alpha_k \left\|\tilde{\mathbf{H}}^l_{RIS,k}[i]\right\|}{\gamma^2} + p^l(x[i])\right] \\ &\quad - \sum_{k=1}^{K} \lambda_k \alpha_k p^l(x[i]) + \sum_{k=1}^{K} \left[\mu_k \alpha_k p^l(x[i]) - P^l_{\text{sum}}(X)\right]. \end{aligned} \tag{22}$$

According to the KKT conditions, in this optimization problem, the complementary slackness condition is given by

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Algorithm 1: PAO Method With RLP Strategy.

Input: k, $p^l(x[i])$, $\tilde{\mathbf{H}}^l_{RIS,k}[i]$, α_k, μ, λ_k
$\mathcal{K} = \{1, \ldots, k, \ldots, K\}$
Define $L_k = \frac{\alpha_k \|\tilde{\mathbf{H}}^l_{RIS,k}[i]\|}{\gamma^2}$
Assume $1/\mu = [p^l(x[i]) + \sum_{k\in K} L_k] / \sum_{k\in K} \alpha_k$
then $\hat{k} = \arg\max_{k\in\mathcal{K}} \frac{L_k}{\alpha_k}$
  while $\frac{L_{\hat{k}}}{\alpha_{\hat{k}}} \geq \frac{1}{\mu}$ **do**
    $\mathcal{K} = \mathcal{K} - \hat{k}$
  end while
Output: $p^l_k(x[i]) = \frac{1}{(\mu-\lambda_k)\alpha_k \cdot \ln 2} - L_k$
Input: $P_k = \{p^l_1(x[1]), \ldots, p^l_K(x[I-1])\}$
Data processing block (BO $\|$ CNN):
BO utilizes Gaussian Processes to select sample points $\tilde{x}[m]$ in **Algorithm 2**.
CNN extracts data characteristics as $\hat{p}^{c_n}_{c_m}$ in **Algorithm 3**.
Prediction block:
  LSTM $\rightarrow$ input gate i^l_t, forget gate f^l_t, output gate O^l_t
  Optimizer block:
    while $\Lambda[t], V[t] \leftarrow \nabla f(\omega[t])$ **do**
      $\omega[t+1] = \omega[t] - \beta \frac{\Lambda[t]}{V[t]}$
    end while
Output: p^l_{pre}

$\lambda_k \alpha_k p^l(x[i]) = 0$, and the stationarity condition is:

$$\frac{\partial \mathcal{L}}{\partial p^l(x[i])} = -\frac{1}{\ln 2} \cdot 1 / \left[\frac{\alpha_k \left\| \tilde{\mathbf{H}}^l_{RIS,k}[i] \right\|}{\gamma^2} + p^l(x[i]) \right] - \lambda_k \alpha_k + \mu_k \alpha_k = 0. \tag{23}$$

Therefore, the following formula can be derived through the deformation and simplification of (23):

$$\left\{ \mu_k \alpha_k - \lambda_k \alpha_k - \frac{1}{\ln 2} \cdot 1 / \left[\frac{\alpha_k \left\| \tilde{\mathbf{H}}^l_{RIS,k}[i] \right\|}{\gamma^2} + p^l(x[i]) \right] \right\} \times p^l(x[i]) = 0. \tag{24}$$

When $\mu_k \geq \gamma^2 / \left(\alpha_k \left\| \tilde{\mathbf{H}}^l_{RIS,k}[i] \right\| \right)$ holds, $p^l(x[i]) = 0$ is inevitable, and when $\mu_k < \gamma^2 / \left(\alpha_k \left\| \tilde{\mathbf{H}}^l_{RIS,k}[i] \right\| \right)$ is established, the transmission power allocated to the k-th reflector element is obtained:

$$p^l_k(x[i]) = \frac{1}{(\mu_k - \lambda_k)\alpha_k \cdot \ln 2} - \frac{\alpha_k \left\| \tilde{\mathbf{H}}^l_{RIS,k}[i] \right\|}{\gamma^2}. \tag{25}$$

B. Data Preprocessing Block

1) BO Method: The Bayesian optimized data can be trained and predicted by LSTM to obtain higher accuracy and increase the accuracy of the proposed RLP method. The data obtained after the initialization of the existing data $(x^l[i], p^l[i])$ is set as $(\tilde{x}[m], f(\tilde{x}[m]))$, a set of hyperparameter models $\mathcal{D}$. Defines the problem of optimal transmission power of RLP system for BO method [29] as:

$$\max_{\tilde{x}\in\mathcal{X}} \quad f(\tilde{x}) \sim \mathcal{GP}(\mu(\tilde{x}), \gamma(\tilde{x}, \tilde{x}')), \tag{26a}$$

$$\text{s.t.} \quad \mathcal{X} \subset \mathbb{R}^d, \tag{26b}$$

$$d \leq d^*, \tag{26c}$$

$$\sum_m \tilde{x}[m] = 1, \quad m = 1, 2, \ldots, M, \tag{26d}$$

where M and d denote the number of hyperparameters, $\tilde{x}[m]$ indicates m-th hyperparameters, the times of evaluation in which the maximum value is d^*. $f(\tilde{x})$ denotes the objective function to be optimized, while $\mathcal{GP}(\cdot,\cdot)$ signifies the Gaussian process, $\mu(\tilde{x})$ serves as the mean function, and $\gamma(\tilde{x}, \tilde{x}')$ acts as the covariance function, depicting the correlation between distinct points $\tilde{x}$ and $\tilde{x}'$. $\mathcal{X}$ denotes the hyperparameter exploration space, i.e., the range used to search for the optimal hyperparameters. In the process of the BO-LSTM data processing method, the main process of BO is divided into Gaussian process regression (GPR) and acquisition function (AF) where the decision rule function of GPR is expressed as:

$$r_{\text{opt}} \mid \tilde{x}[m]_* = \arg\min_{r_{\text{pred}}} \int P\left(r_* \mid \tilde{x}[m]_*, \tilde{X}, r\right) \times \mathcal{L}\left(r_*, r_{\text{pred}}\right) dr_*, \tag{27}$$

for which r_* is the global maximum value and r is the true value vector that has been evaluated. r_{pred} indicate the assumption value. The global maximum value can be written as $\tilde{x}[m]_*$, while $\tilde{X} = \tilde{x}[m]$ is the collection of initialized samples. The function value obtained by GPR is denoted as:

$$f_{\max}(\tilde{x}[m]) \sim \mathcal{N}\left(\mu_m(\tilde{x}[m]), \gamma^2_m(\tilde{x}[m])\right), \tag{28}$$

where $\mu_m(\tilde{x}[m])$ and $\gamma^2_m(\tilde{x}[m])$ express the mean and variance functions corresponding to the distributions that the values of the functions obtained from any $\tilde{x}$ to GPR decision rule. Based on the Gaussian distribution, the existing parameter distribution is expressed as:

$$P(\tilde{x}[m] \mid \gamma_m(\tilde{x}[m])) = \frac{1}{\sqrt{2\pi}} \exp\left\{ -\frac{[\tilde{x}[m] - \gamma_m(\tilde{x}[m])]^2}{2} \right\}. \tag{29}$$

Using the joint probability distribution of parameters and data:

$$P\left(\tilde{X}, \gamma_m(\tilde{x}[m])\right) = P(\gamma_m(\tilde{x}[m])) \prod_{m=1}^{M} P(\tilde{x}[m] \mid \gamma_m(\tilde{x}[m])). \tag{30}$$

All the information about $\gamma_m(\tilde{x}[m])$ obtained by $\tilde{X}$ can be summarized as the posterior distribution:

$$P\left(\gamma_m(\tilde{x}[m]), \tilde{X}\right) = \frac{P\left(\tilde{X}, \gamma_m(\tilde{x}[m])\right)}{P(\tilde{X})}$$

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Algorithm 2: BO-LSTM Assisted RLP Method.

1: **Input**: $f, \mathcal{X}, \alpha_m, \mathcal{D}$ $\quad \mathrm{P}^l \leftarrow \mathrm{InitSamples}(f, \mathcal{X})$
2: **for** $m \leftarrow |\mathrm{P}^l|$ to d do
$P_r(y \mid \tilde{x}, \mathrm{P}^l) \leftarrow FitModel(\mathcal{D}, P^l)$
$\tilde{x}[m] \leftarrow \arg\max_{\tilde{x}\in\mathcal{X}} \alpha_m(\tilde{x}, P(y \mid \tilde{x}, \mathrm{P}^l))$
$y[m] \leftarrow f(\tilde{x}[m]) \quad \mathrm{P}^l_m \leftarrow$
$\mathrm{P}^l \cup (\tilde{x}[1], y[1]) \cup (\tilde{x}[2], y[2]) \cup \ldots \cup (\tilde{x}[m], y[m])$
3: The forget gate control the data which need to be ignored:

$$f_t^l = \sigma\left(W_f^l \cdot \left[h_{t-1}^l, \hat{x}_t^l\right] + b_f^l\right)$$

4: The output of forget gate at time t is obtained, then the relevant information of $\hat{x}_t^l$ is retained:

$$i_t^l = \sigma\left(W_i^l \cdot \left[h_{t-1}^l, \hat{x}_t^l\right] + b_i^l\right)$$

$$\tilde{C}_t^l = \tanh\left(W_c^l \cdot \left[h_{t-1}^l, \hat{x}_t^l\right] + b_c^l\right)$$

5: With the forget gate and output gate, the values of the updated nerve cells can be obtained.

$$C_t^l = f_t^l * C_{t-1}^l + i_t^l * \tilde{C}_t^l \leftarrow \tilde{C}_t^l$$

6: Using the output gate control C_t^l output, and through O_t^l control, get the final h_t^l.

$$O_t^l = \sigma\left(W_o^l \cdot \left[h_{t-1}^l, \hat{x}_t^l\right] + b_o^l\right)$$

$$h_t^l = O_t^l * \tanh\left(C_t^l\right) \leftarrow C_t^l$$

7: **End for**

$$= \frac{P\left(\gamma_m(\tilde{x}[m])\right) \prod_{m=1}^{M} P\left(\tilde{x}[m] \mid \gamma_m(\tilde{x}[m])\right)}{\int P\left(\gamma_m'(\tilde{x}[m])\right) \prod_{m=1}^{M} P\left(\tilde{x}[m] \mid \gamma_m(\tilde{x}[m])\right) d\gamma_m'(\tilde{x}[m])}$$

$$= \frac{1}{\sqrt{2\pi\sigma_M^2}} \exp\left\{-\frac{1}{2\sigma_M^2}\left[\gamma_m(\tilde{x}[m]) - \hat{\gamma}_M(\tilde{x}[m])\right]^2\right\}, \quad (31)$$

and

$$\log P\left(\gamma_m(\tilde{x}[m]), \tilde{X}\right) = \log P\left(\gamma_m(\tilde{x}[m])\right) + \sum_{m=1}^{M} \log P\left(\tilde{x}[m] \mid \gamma_m(\tilde{x}[m])\right) + Constant, \quad (32)$$

where

$$\hat{\gamma}_M(\tilde{x}[m]) = \frac{1}{M+1}\sum_{m=1}^{M} \tilde{x}[m], \quad \sigma_M^2 = \frac{1}{M+1}. \quad (33)$$

It is assumed that the true distribution finally obtained is q. As long as the mean and variance of q are finite, the pseudo-distance between q and the parameter distribution $p(\cdot \mid \gamma_m(\tilde{x}[m]))$ is measured by calculating Kullback-Leibler Divergence (KL-D) and is expressed as:

$$\begin{aligned} &D_{KL}\left[q \| P\left(\cdot \mid \gamma_m(\tilde{x}[m])\right)\right] \\ &= -\mathbb{E}_q\left[\log P\left(\tilde{X} \mid \gamma_m(\tilde{x}[m])\right)\right] + Constant \\ &= \mathbb{E}_q\left[\frac{\left(\tilde{x}[m] - \gamma_m(\tilde{x}[m])\right)^2}{2}\right] + Constant \\ &= \frac{1}{2}\left[\gamma_m(\tilde{x}[m]) - \mathbb{E}_q[\tilde{X}]\right]^2 + Constant. \end{aligned} \quad (34)$$

When $p(\cdot \mid \gamma_m(\tilde{x}[m]))$ approaches q infinitely, the following distribution can be obtained:

$$\begin{aligned} &P^*\left(\gamma_m(\tilde{x}[m])\right); M) \\ &= \frac{\exp\left\{-M \cdot D_{KL}\left[q \| P\left(\cdot \mid \gamma_m(\tilde{x}[m])\right)\right]\right\}}{Z} \\ &= \frac{\exp\left\{-\frac{M}{2}\left[\gamma_m(\tilde{x}[m]) - \mathbb{E}_q[\tilde{X}]\right]^2\right\}}{Z} \\ &= \frac{1}{\sqrt{2\pi\left(\sigma_M^*\right)^2}} \exp\left\{-\frac{1}{2\left(\sigma_M^*\right)^2}\left[\gamma_m(\tilde{x}[m]) - \gamma_m^*(\tilde{x}[m])\right]^2\right\}, \end{aligned} \quad (35)$$

where

$$\gamma_m^*(\tilde{x}[m]) = \mathbb{E}_q[\tilde{X}], \quad \left(\sigma_M^*\right)^2 = \frac{1}{M}, \quad (36)$$

Z is the normalized constant. And because of $\lim_{M\to\infty} \hat{\gamma}_M(\tilde{x}[m]) = \gamma_m^*(\tilde{x}[m]), \lim_{M\to\infty} \frac{\sigma_M^*}{\sigma_M} = 1$, the verification shows that $P(\gamma_m(\tilde{x}[m]), \tilde{X})$ is concentrated around the mean of q.

According to the symmetry of normal distribution of the BO method in this section, AF chooses probability of improvement as the optimization method and is expressed as:

PI (Probability of Improvement):

$$\begin{aligned} &\arg\max \alpha_m(\tilde{x}) = \arg\max P_r\left(f(\tilde{x}) \geq \tilde{r}_* + \varepsilon\right) \\ &= \arg\max P_r\left(\frac{f(\tilde{x}) - \mu_m(\tilde{x})}{\gamma_m(\tilde{x})} \leq \frac{\tilde{r}_* + \varepsilon - \mu_m(\tilde{x})}{\gamma_m(\tilde{x})}\right) \\ &= \arg\max P_r\left(\frac{f(\tilde{x}) - \mu_m(\tilde{x})}{\gamma_m(\tilde{x})} \geq \frac{\mu_m(\tilde{x}) - \tilde{r}_* - \varepsilon}{\gamma_m(\tilde{x})}\right) \\ &= \arg\max \Phi\left(\frac{\mu_m(\tilde{x}) - \tilde{r}_* - \varepsilon}{\gamma_m(\tilde{x})}\right), \end{aligned} \quad (37)$$

where $\tilde{r}_* = \max_{j\leq m} f(\tilde{x}[j])$ with $j = 1, 2, \ldots, m$, denotes the local maximum value of function, while ε is used to adjust the hyperparameters for the next exploration. $\Phi(\cdot)$ and $P_r(\cdot)$ denote the cumulative distribution function and probability density function of statistics, respectively.

2) CNN Process: When using CNN architecture [17] to be the beginning of the data processing, it is utilized to extract the time series data $p^l[i]$ on the spatial structure of the relationship. The main function of CNN network architecture is to extract the main data features, improve the generalization performance, and speed up the training speed. These functions are realized by the convolutional layer and pooling layer in CNN. For the time series data in this paper, a one-dimensional CNN has been used for processing, and the c_m-th convolutional map of the c_n layer

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Algorithm 3: CNN-LSTM Assisted RLP Method.

1: **Begin**
2: The input data P^l is preprocessed by CNN to extract the characteristic values of the data and reduce the data volume.
3: Softmax layer serves as the output layer and converts the output of the convolutional layer into a vector. The function corresponding to softmax is expressed as:

$$\sigma(z) = \frac{e_A^z}{\sum_{a=1}^{A} e_a^z},$$

where z is the output vector of the fully connected layer, denotes the number of neurons in the output layer.
4: The data processed by CNN is brought into the LSTM as the input data $\hat{x}_t^l$.
5: Through the control of LSTM input gate, output gate and forget gate mentioned in Algorithm 2, the required transmission power prediction data p_{pre}^l is obtained.
6: **End**

obtained is expressed as follows:

$$\tilde{p}_{c_m}^{c_n} = \mathcal{F}\left(\sum_{z=1}^{N} \tilde{p}_i^{c_n-1} * w_{i,c_m}^{c_n} + \hat{b}_{c_m}^{c_n}\right), \tag{38}$$

where $\mathcal{F}(\cdot)$ is the activation function, $w_{i,c_m}^{c_n}$ is the weight of the c_m-th convolution kernel of the c_n layer, $\hat{b}_{c_m}^{c_n}$ is the bias of the c_m-th convolution kernel of the c_n layer, and n is the number of input convolution maps. This algorithm uses the max pooling method to process sequence data. The maximal feature sequence obtained by pooling is denoted by

$$\hat{p}_{c_m}^{c_n} = \max\left(\tilde{p}_{c_m}^{c_n} : \tilde{p}_{c_m+\star-1}^{c_n}\right), \tag{39}$$

where $\star$ is the number of pooling windows. The obtained maximum feature sequence can reduce the amount of data under the premise of better retaining features. CNN's loss function is the core part of structure risk function, which is expressed as:

$$\Gamma^* = \arg\min_{\Gamma} \frac{1}{I}\sum_{i=0}^{I-1} \underbrace{L_{CNN}\left\{p^l[i], f(x[i];\Gamma)\right\}}_{\text{Loss function}} + \underbrace{\lambda_R R(\Gamma)}_{\text{Regularizer}}, \tag{40}$$

where

$$L_{CNN}\left\{p^l[i], f(x[i];\Gamma)\right\} = \frac{1}{I}\sum_{i=0}^{I-1} \exp\left\{-p^l[i] f(x[i];\Gamma)\right\}, \tag{41}$$

Γ stand for parameters that need to be regularized in the CNN model, including weights and biases. $R(\cdot)$ and λ_R denote regular function and regularization strength, respectively.

C. LSTM Prediction Block

The processed data sequence is used as the LSTM input data $\hat{x}_t^l$ for training, and the trained prediction model is used to predict the UE's position and power information backward. The hidden layer of an LSTM contains input gates i_t^l, forget gates f_t^l, output gates O_t^l, and cells to store the memory [7]. When the input is trained by the LSTM network, $\hat{x}_t^l$ goes through the nonlinear transformation of the input gate, and then overlaps with the memory cell state processed by the forget gate to form a new memory cell state.

The value of the output signal of the forgetting gate f_t^l determines the forgetting ratio of the memory unit, which determines the information to be forgotten. For example, when the UE's position changes a small distance and the transmission power changes too fast, the data generated will be determined as the data to be forgotten. $\tilde{C}_t^l$ point multiplies the value of the prepared information input into the memory cell with i_t^l to obtain the information in the memory cell.

The transmission power prediction data required by BS at different locations in the proposed RLP system obtained by LSTM network is expressed as follows:

$$p_{\text{pre}}^l = \sigma\left(W_o^l \cdot \left[h_{t-1}^l, \hat{x}_t^l\right] + b_o^l\right), \tag{42}$$

where W_o^l and b_o^l indicate the neuron threshold and weight parameter in the final trained model. The implicate state is denoted as h_{t-1}^l. Building upon the advantages of traditional LSTM, an improved LSTM algorithm is proposed as shown in Fig. 6. This method first models the time slot power sequence data using a bi-directional LSTM (BiLSTM) network composed of forward LSTM and backward LSTM, which $\{h_{t,L0}, h_{t,L1}, \ldots, h_{t,Ln}\}$ and $\{h_{t,R0}, h_{t,R1}, \ldots, h_{t,Rn}\}$ are hidden layer state sequences of forward LSTM and backward LSTM. Then, it utilizes Adaboost to weight-combine the predictions of the BiLSTM network. Adaboost initially sets the initial weight distribution to

Algorithm 4: BiLSTM-Adamboost Prediction Strategy for RLP Method.

1: **Begin**
2: The weight distribution of each sample in the training data is initialized as a uniform distribution:
3: $D_1^{(i)} = (\omega_0, \ldots, \omega_i, \ldots, \omega_{I-1}) = (\frac{1}{I}, \ldots, \frac{1}{I}, \ldots, \frac{1}{I})$
4: **for** $t = 1, \ldots, T$ **do**
5: The input sequence undergoes bidirectional sequence modeling through BiLSTM, bidirectional hidden state sequences are generated:
6: $\{h_{t,L0}, h_{t,L1}, \ldots, h_{t,Ln}\}$ and $\{h_{t,R0}, h_{t,R1}, \ldots, h_{t,Rn}\}$.
7: The prediction output is obtained through a fully connected layer: $\hat{p}_{\text{t,predict}}^l(x[i])$.
8: Calculate the error rate of the current predictor H_t:
9: $\varepsilon_t = P\{H_t(x[i]) \neq p^l(x[i])\}$.
10: Calculate the weight of the current predictor:
11: $\alpha_t = \frac{1}{2}\ln(\frac{1-\varepsilon_t}{\varepsilon_t})$.
12: Update the distribution of sample weights:
13: $D_{t+1} = D_t(i)\exp[-\alpha_t \cdot p^l(x[i])H_t(x[i])]/Z_t$,
14: where $Z_t = 2\sqrt{\varepsilon_t(1-\varepsilon_t)}$ is normalization function.
15: Output the final ensemble predictor:
16: $H_f = \text{sign}(\sum_{t=1}^{T} \alpha_t H_t(x))$.
17: **end for**

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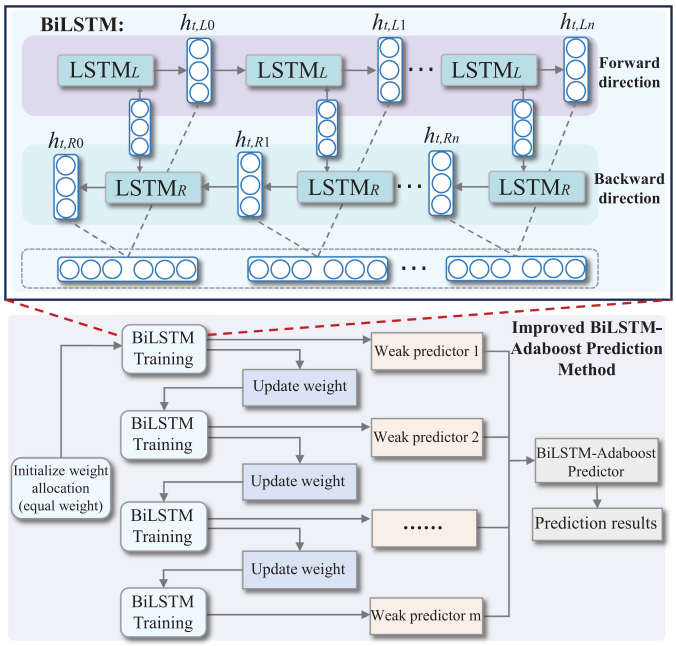

Fig. 6. Network architecture of improved BiLSTM-Adaboost prediction algorithm for proposed localization model.

be equal, and then iteratively updates the weight distribution. In each weight update iteration of Adaboost, a weak predictor is trained using BiLSTM, and the final prediction of the time slot power sequence data p^l_{predict} is obtained based on the collective performance of all weak predictors.

D. Optimizer Block

Adam and SGDM optimizer, which have different characteristics, for different functions and different parameter settings will have the different effects of prediction accuracy of transmit power data. Adam optimizer is suitable for a variety of DL tasks and data sets, while SGDM optimizers are generally suitable for large data sets and complex deep neural networks.

1) SGDM: SGDM introduces first-order momentum on the basis of SGD, that is, inertia is added to the process of gradient descent. The descent direction is determined not only by the gradient direction of the current point but also by the previously accumulated descent direction. The first and second order momentum [44] at time t can be represented according to the historical gradient:

$$\Lambda[t] = \phi(\text{grad}[1], \text{grad}[2], \ldots, \text{grad}[t]), \tag{43}$$

$$V[t] = \psi(\text{grad}[1], \text{grad}[2], \ldots, \text{grad}[t]), \tag{44}$$

$$\Lambda_{SGDM}[t] = \mu_1 \cdot \Lambda[t-1] + (1-\mu_1) \cdot \text{grad}[t], \tag{45}$$

which $\text{grad}[t] = \nabla f(w[t])$, $w[t]$ and μ_1 is the parameter needed to be optimized and hyperparameter that controls first-order momentum, $\phi(\cdot)$ and $\psi(\cdot)$ represent sets. The advantage of SGDM is that using momentum can accelerate the training process, and the obstacle is that it will stay at a local minimum without further improving the performance.

2) Adam: Adam [45] adds second-order momentum on the basis of SGDM, which is an optimization method combining first-order momentum and second-order momentum. The second-order momentum is expressed as:

$$V_{Adam}[t] = \mu_2 \cdot V[t-1] + (1-\mu_2) \cdot \text{grad}[t]^2, \tag{46}$$

where μ_2 denotes the hyperparameter that controls second-order momentum. To guarantee the convergence of the algorithm, it is necessary to control the change of the second-order momentum:

$$V_{Adam}[t] = \max(\mu_2 * V[t-1] + (1-\mu_2) \cdot \text{grad}[t]^2, V[t-1]), \tag{47}$$

ensure that: $\|V[t]\| > \|V[t-1]\|$. The descent gradient at the current moment can be calculated as $\eta[t] = \frac{\beta_i \Lambda[t]}{\sqrt{V[t]}}$ and the optimization parameters are updated according to the descending gradient as $w[t+1] = w[t] - \eta[t]$ which β_i symbolize the initial learning rate. Adam optimizer has good adaptability but is prone to overfitting the training set.

Assuming that the prediction data p^l_{predict} is affected by an error ε_p with a mean of 0 and a variance of σ_p, i.e., $p^l_{\text{predict}} = p^l[i] + \varepsilon_p$, the probability density function and Fisher information matrix of the predicted value are represented as:

$$f(p^l_{\text{predict}}[i] \big| p^l[i]) = \frac{1}{\sqrt{2\pi}\sigma_p} \exp\left[-\frac{(p^l_{\text{predict}}[i] - p^l[i])^2}{2\sigma_p^2}\right], \tag{48}$$

$$\begin{aligned} \text{J}_F(p^l[i]) &= -\frac{\partial^2 \ln[f(p^l_{\text{predict}}[i] \big| p^l[i])]}{\partial (p^l[i])^2} \\ &= \frac{\partial^2}{\partial (p^l[i])^2} \frac{(p^l_{\text{predict}}[i] - p^l[i])^2}{2\sigma_p^2} = \frac{1}{\sigma_p^2}, \end{aligned} \tag{49}$$

which $p^l_{\text{predict}}[i]$ denotes the predicted value obtained from the prediction of the UE power data at i-th time slot, respectively. The accuracy of positioning problem is expressed as the CramÃ©r-Rao lower bounds (CRLB):

$$\text{CRLB}_{RLP} \geq \text{J}_F^{-1}(p^l[i]) = \sigma_p^2. \tag{50}$$

V. Simulation Results

This section presents numerical simulation results to demonstrate the effectiveness of the RLP method in a wireless communication scenario. The channel pathloss model assumes the 3GPP Urban Micro (UMi) [46] model with both line-of-sight (LOS) and non-line-of-sight (NLOS) versions at a carrier frequency of 3 GHz. During the LSTM training process, a maximum of 100 iterations were used, and the model was evaluated every 200 epochs. The system requires an achievable rate of 4 and 6 bits/s/Hz, and the main parameters used in the simulations are listed in Table III.

A. Algorithm Performance Analysis

Fig. 7 illustrates the simulation results of the channel capacity variation with respect to SNR for the proposed PAO algorithm in Section IV-A. The simulation shown in Fig. 7 was conducted

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TABLE III
MAIN PARAMETERS AND VALUES OF SIMULATION

LSTM Parameters	Value	Model Parameters	Value
Batch size	512	Achievable rate ζ, [bits/s/Hz]	4 & 6
Initial learn rate	[0.01,0.5]	Amplitude reflection coefficient	1
Number of hidden layers	1	Minimum distance between UE and BS d^l_{BD}, [m]	10
Learn rate drop period	100	Distance between RIS and BS d^l_{SR}, [m]	100
Learn rate drop factor	0.2	Bandwidth, [GHz]	10
Max epochs	200	Carrier frequency f_c, [GHz]	3
Hidden size	[16,64]	Noise figure n, [dB]	10
		Phase adjustment range	[0,2π]

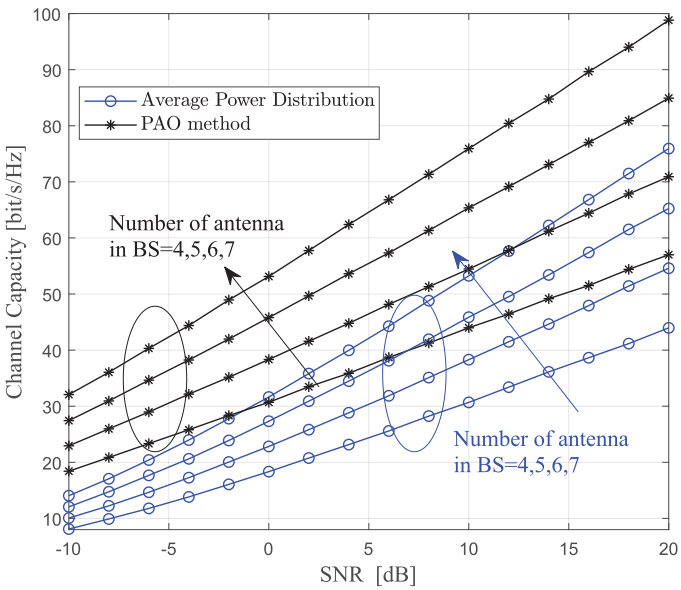

Fig. 7. Channel capacity versus SNR for PAO method compared with average power distribution.

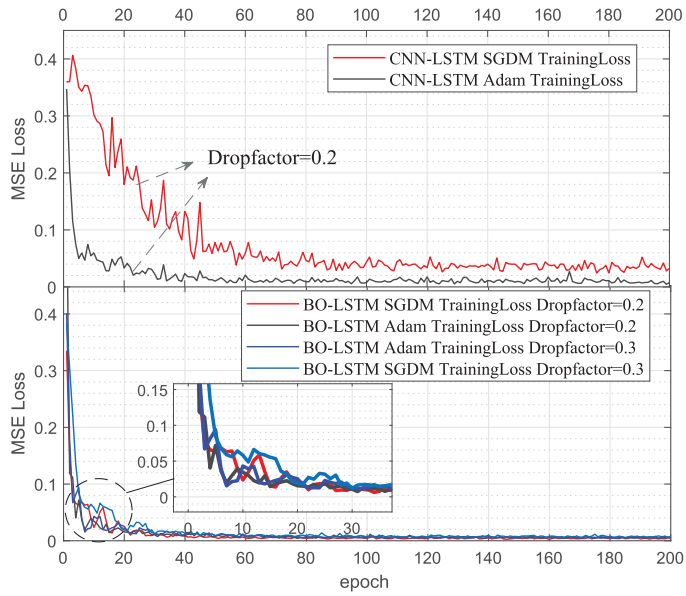

Fig. 8. The MSE loss curve in LSTM training process with evaluation epochs of 200.

in an RIS-assisted MIMO scenario. From the graph, it is evident that the channel capacity obtained by the PAO algorithm significantly outperforms the method of uniformly allocating power to each reflecting element on the RIS. The optimized channel capacity is more conducive to the stable transmission of signals and the collection of data in the signal localization process facilitated by the RLP algorithm. Fig. 8 shows the performance of the loss function during the LSTM training process, which demonstrates that the location model is well-trained. The BO-LSTM loss function under the Adam optimizer quickly converges to zero and outperforms other algorithms. The loss value remains stable from epoch 60 onwards, indicating no training issues occurred during the process. These results confirm that the model weights are well-tuned.

Fig. 9 shows a comparison of the location error calculated by CRLB between the BO-LSTM and CNN-LSTM algorithms using different optimizers, namely Adam and SGDM. The calculations in Figs. 9 and 10 are based on the curves of Fig. 13(a) and the left section of Fig. 13(c), taking the best curve fit results. The results indicate that the proposed improved BO-BiLSTM-Adamboost and CNN-BiLSTM-Adaboost algorithm achieves a higher positioning accuracy. On the other hand, the CNN-LSTM algorithm performs better than BO-LSTM for distances ranging from 75[m] to 100[m]. Additionally, the error calculated by

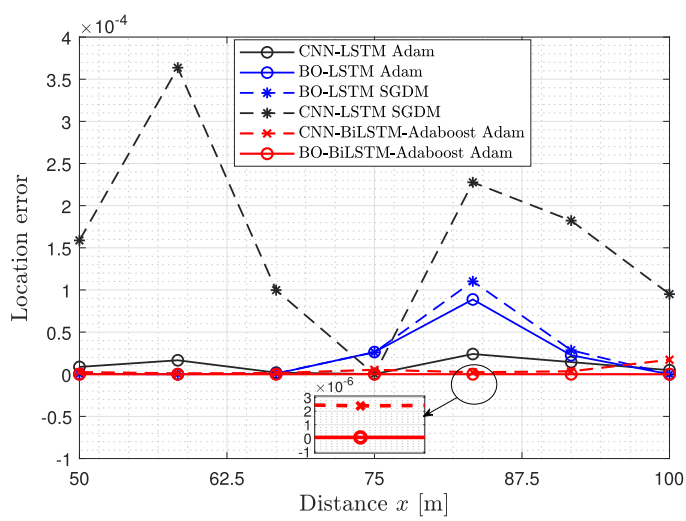

Fig. 9. Location prediction error based on CRLB versus distance x.

CRLB reaches 6.5×10^{-6} and the accuracy of the BO-LSTM algorithm for localization prediction reaches 96.86% in the first half of the prediction distance according to the calculated result of average related error. Table IV presents a comparison of localization accuracy achieved by LSTM, machine learning techniques, and user classification methods in different studies.

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TABLE IV
THE LOCALIZATION ACCURACY IN DIFFERENT METHOD OF REFERENCE

Reference	Parameters setting	Main content	Algorithm & Accuracy
This paper	Shown in Table III	Localization Prediction for RIS-aided System	BO-LSTM & 96.8%
[7]	Not mention	Mobility Prediction	LSTM & 92.5%
[8]	Number of input: 2, Number of output: 1, Number of epoch: 245, Number of sample: 7000.	Network Location Prediction	Machine Learning & 94.3%
[9]	Input and attention layer: 64, Hidden layer: 64, Hidden layer dimension: 128, Learning rate: 0.001.	Surrounding Vehicle Trajectory Prediction	LSTM & 91%
[18]	Input layer: 1, Hidden layer: 3, Output layer: 1.	Position Prediction Model for Mobile Users	LSTM & 90%
[47]	Number of samples: 10000, Hidden layer: 500, Initial learning rate: 0.0005.	Location Prediction	LSTM-LSTM & 64.49%

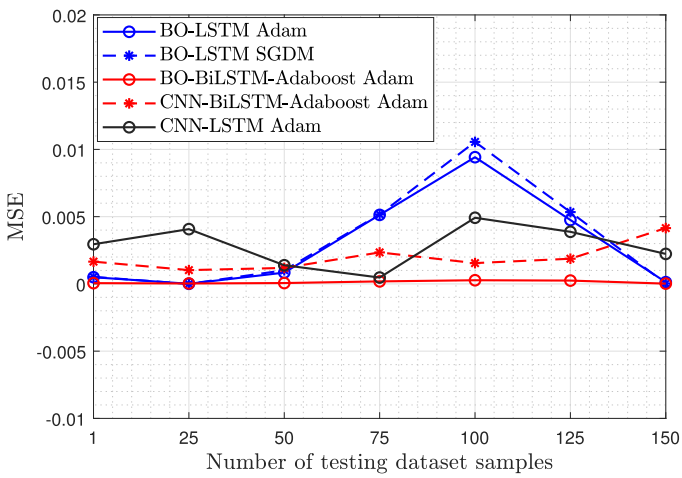

Fig. 10. MSE versus varying number of testing dataset samples in different algorithms.

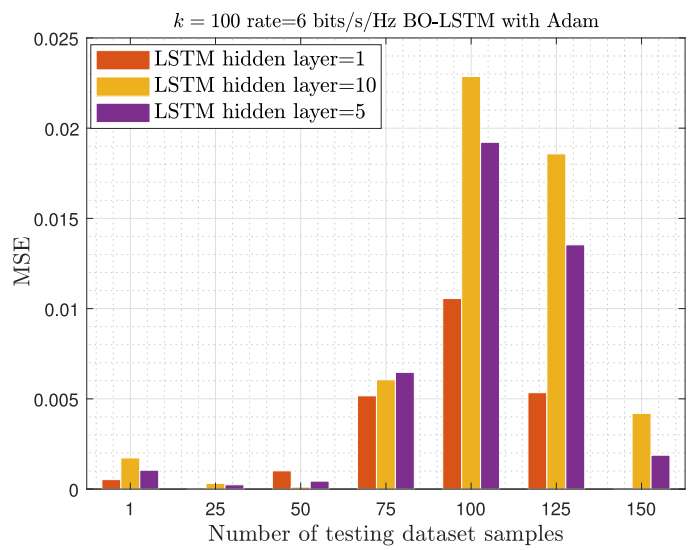

Fig. 11. The MSE for different number of LSTM hidden layer.

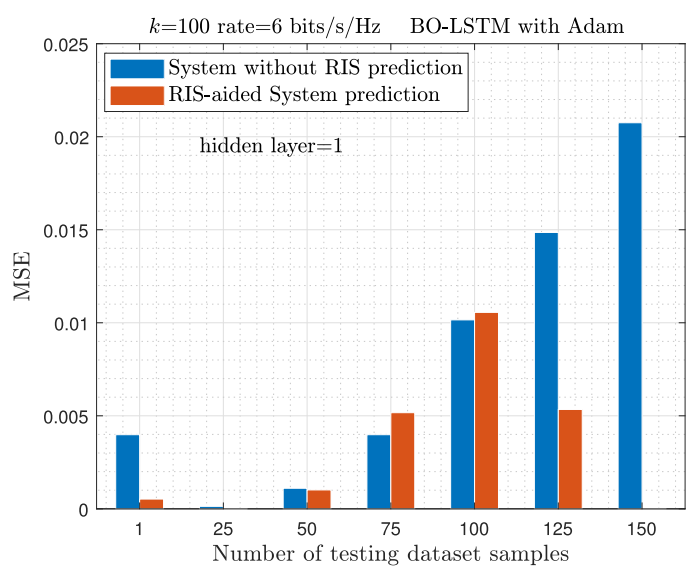

Fig. 12. MSE comparison of localization predictions for systems with and without RIS assistance.

The effectiveness of the Adam optimizer is more pronounced in CNN-LSTM, while in BO-LSTM, the Adam optimizer shows slightly better performance than SGDM. Fig. 10 compares the performance of BO-LSTM and CNN-LSTM algorithms in terms of mean squared error (MSE) under different optimizers. MSE is computed as $MSE = \frac{1}{I}\sum_{i=0}^{I-1}(p^l_{\text{predict}}[i] - p^l_{\text{actual}}[i])^2$, which $p^l_{\text{actual}}[i]$ denotes the actual value of the transmit power data in the i-th time slot at the l-th position. As shown in Fig. 10, the MSE of BO-LSTM remains stable between 0 to 0.01 under both optimizers, indicating its robustness. On the other hand, the MSE performance of CNN-LSTM under SGDM optimizer is worse than that of under Adam, which could be attributed to the fact that adaptive learning rate algorithms such as Adam work better for sparse data. Furthermore, a comparison is made between the proposed enhanced models BO-BiLSTM-Adaboost and CNN-BiLSTM-Adaboost in terms of MSE performance under the Adam optimizer. The results from the graph show a clear improvement in the performance of BO-BiLSTM-Adaboost, which surpasses that of CNN-BiLSTM-Adaboost.

Fig. 11 shows the impact of the number of LSTM hidden layers on the prediction performance of BO-LSTM using the Adam optimizer. The results show that the best error performance is achieved when the number of hidden layers is set to 1, and worse performance is obtained with 5 or 10 hidden layers. This suggests that a simpler model is more suitable for modeling and prediction when using the BO-LSTM algorithm. Fig. 12 compares the positioning accuracy of the system with and without the assistance of RIS. The results show that the RIS-assisted wireless communication system achieves higher

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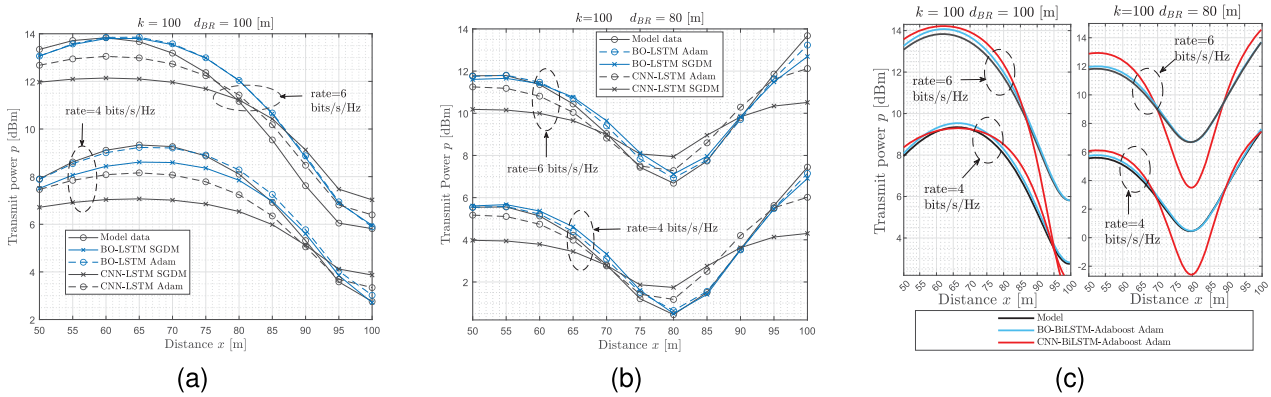

Fig. 13. The transmit power needed for system versus moving distance of UE in a single antenna scenario. (a) The prediction result comparison for a single antenna scenario using BO-LSTM and CNN-LSTM algorithms under the Adam and SGDM optimizers when $d_{BR} = 100$ [m]. (b) The prediction result comparison for a single antenna scenario using BO-LSTM and CNN-LSTM algorithms under the Adam and SGDM optimizers when $d_{BR} = 80$ [m]. (c) The prediction result comparison for a single antenna scenario using BO-BiLSTM-Adaboost and CNN-BiLSTM-Adaboost algorithms under Adam optimizer when $d_{BR} = 100$ [m] and $d_{BR} = 80$ [m].

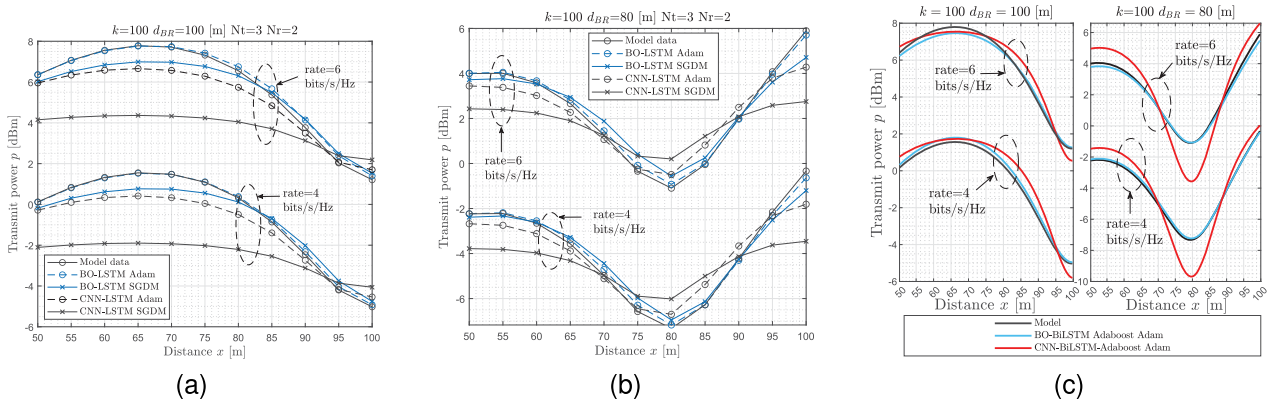

Fig. 14. Transmit power needed for system versus moving distance of UE in a MIMO setting. (a) The prediction result comparison for a MIMO setting using BO-LSTM and CNN-LSTM algorithms under the Adam and SGDM optimizers when $d_{BR} = 100$ [m]. (b) The prediction result comparison for a MIMO setting using BO-LSTM and CNN-LSTM algorithms under the Adam and SGDM optimizers when $d_{BR} = 80$ [m]. (c) The prediction result comparison for a MIMO setting using BO-BiLSTM-Adaboost and CNN-BiLSTM-Adaboost algorithms under Adam optimizer when $d_{BR} = 100$ [m] and $d_{BR} = 80$ [m].

positioning accuracy and lower error rate than the system without RIS. This is because the data structure obtained by the RIS-assisted wireless communication system is more suitable for the algorithm, leading to better positioning accuracy.

B. Simulation Results of Proposed RLP Method

Fig. 13(a) displays the predicted relationship between the distance x of the UE from the origin and the required transmission power p for the system, using the BO-LSTM and CNN-LSTM algorithms under the Adam and SGDM optimizers, for a single antenna setting. The curves were obtained by simulating the prediction with 100 RIS elements. The results demonstrate that the BO-LSTM algorithm, when coupled with the Adam optimizer, exhibits a strong alignment with the model data, as derived in formula (10) for $\zeta = 4$ and 6 bits/s/Hz. On the other hand, the SGDM optimizer outperforms when ζ is set to 6 bits/s/Hz. The curves show that the maximum required transmission power of the system is attained at a distance of 60[m] when $\zeta = 6$ bits/s/Hz, and at a distance of 65[m] when $\zeta = 4$ bits/s/Hz. Fig. 13(b) shows the curve prediction of the transmission power when the distance between the BS and RIS is $d_{BR} = 80$[m]. Using this information, the location of the UE and the extreme point of the transmission power can be predicted, based on the value of the required transmission power for the system. The curves in Fig. 13(c) illustrate the fitting performance of the proposed improvement algorithm based on BO-BiLSTM-Adaboost and CNN-BiLSTM-Adaboost using the Adam optimizer at $d_{BR} = 100$[m] and $d_{BR} = 80$[m]. It is evident from the graph that the combination of BO with BiLSTM-Adaboost achieves better fitting results.

When it comes to the practical application of 6G wireless communication, the main application scenario for RIS is multi-antenna communication systems. Therefore, in Fig. 14, a comparison was made between different algorithms in terms of their fitting performance with respect to transmission power and location in a MIMO setting. From Fig. 14(a) and (b), it can be observed that when the number of transmitting antennas is set to 3 and the number of receiving antennas is set to 2,

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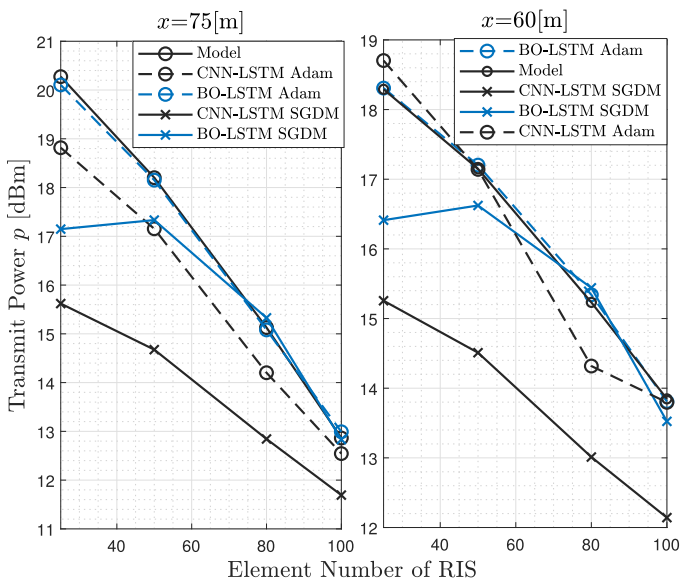

Fig. 15. Transmit power needed with respect to the number of RIS elements at two different locations of UE, with an achievable rate $\zeta = 6$ bits/s/Hz.

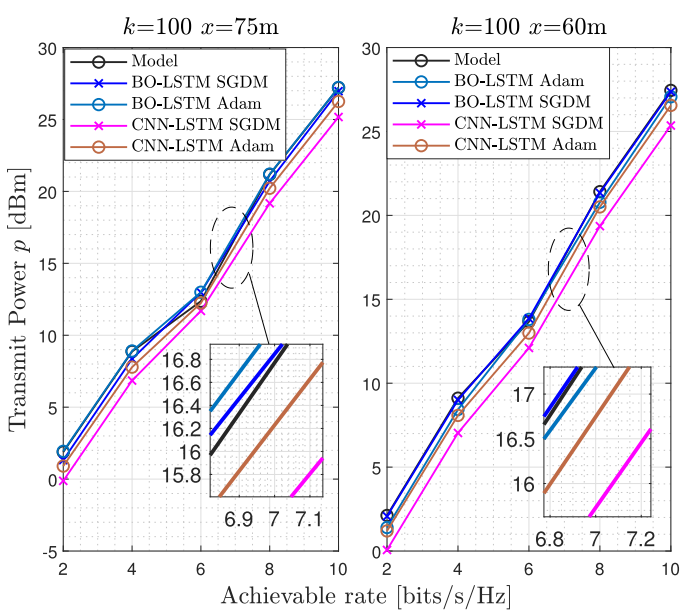

Fig. 16. Transmit power versus varying achievable rate under different algorithms at two different locations of UE, with the element number of RIS k=100.

BO-LSTM with Adam still achieves the best fitting performance compared to other algorithms. Furthermore, from the results of the fitting curves graph in Fig. 14(c), it can be seen that BO-BiLSTM-Adaboost has also achieved better performance. The simulation results in Fig. 14 further validate the conclusions obtained in Fig. 13, indicating that the RLP algorithm remains applicable in multi-antenna communication scenarios. A comparative analysis of the fitting prediction for different achievable rates and numbers of RIS elements is performed when the UE distance x is at the extreme point ($x_1 = 60$[m]) and non-extreme point ($x_2 = 75$[m]) in Fig. 16 of this section.

For the two different positions $x_1 = 60$[m] and $x_2 = 75$[m] of UE, the changes in the transmission power required by the system under different numbers of RIS elements and the predictions of four different algorithms are shown in Fig. 15. From the perspective of the prediction of different algorithms, the BO-LSTM algorithm using Adam optimizer performs relatively well in two different positions, which indicates that Adam optimizer has a better prediction effect in the adjustment of RIS parameters for the prediction of UE. The trend of the required transmission power of the system for different achievable rates is shown in Fig. 16. The comparison of the prediction situation of the UE at two different locations is depicted. It can be concluded from the fitting situation predicted by different algorithms that the BO-LSTM algorithm under the condition of both Adam and SGDM optimizer has a better performance than that of CNN-LSTM algorithm.

VI. Conclusion

This paper considered a wireless communication localization scenario based on a RIS for auxiliary support. Using prior data on the UE's performance, a training model was developed to forecast the relationship between UE location and transmit power. Time series prediction algorithms, namely BO-LSTM and CNN-LSTM, were compared for accuracy, with one algorithm based on BO data processing method and the other on CNN for feature extraction. Both methods provided accurate positioning predictions, with BO-LSTM optimized by Adam outperforming the other algorithms.

Compared to other neural networks, LSTM can effectively process small amounts of data for prediction, laying the foundation for future communication positioning and UE performance prediction. Moreover, if the modeling prediction of one distance is considered as a unit distance in a long distance, and the long distance is divided into countless unit distances, continuous long-term prediction can be made with further verification in real-time. As the UE's transmit power and other performance may be similar in some location points, the predicted location information needs to be considered and corrected based on the distance constraint range.

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